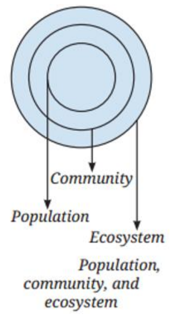
DELHI WORLD PUBLIC SCHOOL, AJMER
SUBJECT- SCIENCE
CLASS- VIII
Chapter -12 How Nature Work In Harmony

Question 1. Refer to the given diagram (Fig.) and select the wrong statement.



(i) A community is larger than a population.
(ii) A community is smaller than an ecosystem.
(iii) An ecosystem is part of a community.
Answer:
(iii) An ecosystem is part of a community.

Question 2. A population is part of a community. If all decomposers suddenly disappeared from a forest ecosystem, what changes do you think would occur? Explain why decomposers are essential.
Answer: Changes: Dead plants and animals would accumulate, nutrients would not return to the soil, and plant growth would decline. This could lead to a reduction in herbivore and carnivore populations due to food scarcity.
Why Essential: Decomposers (e.g., fungi, bacteria) break down dead matter into simpler substances, recycling nutrients like nitrogen and carbon back into the soil for plants. Without them, the nutrient cycle would stop, disrupting the food web.

Question 3. Selvam from Cuddalore district, Tamil Nadu, shared that his milage was less affected by the 2004 Tsunami compared to nearby milages due to the presence of mangrove forests. This surprised Sarita, Shabnam, and Shijo. They wondered if mangroves were protecting the village. Can you help them understand this?
Answer: Yes, mangroves protected the village. Mangrove forests act as natural barriers, slowing down strong winds and waves during storms and tsunamis. Their roots stabilize soil, reducing erosion, and they absorb water impact, protecting coastal areas. The Sundarbans' mangroves, a World Heritage Site, demonstrate this protective role.

Question 4. Look at this food chain: Grass → Grasshopper → Frog → Snake
If frogs disappear from this ecosystem, what will happen to the population of grasshoppers and snakes? Why?
Answer:
- **Grasshoppers: The population would likely increase because frogs, their main predators, are gone, reducing predation pressure.**
- **Snakes: The population would decrease because frogs are their primary food source, leading to food scarcity.**
- **Why: In a food chain, the removal of a middle trophic level (frogs) disrupts the balance. Fewer predators allow prey (grasshoppers) to multiply, while the loss of prey affects higher predators (snakes).**

Question 5. In a school garden, students noticed fewer butterflies the previous season. What could be the possible reasons? What steps can students take to have more butterflies on campus?
Answer:

- **Possible reasons: Possible causes include pesticide use killing larvae, loss of nectar plants, or increased predators (e.g., birds). Monoculture or pollution might also reduce habitat quality.**

- **The steps we can take: Plant diverse nectar-rich flowers (e.g., marigolds), avoid pesticides, create sheltered areas, and install butterfly feeders or water sources to attract and support butterflies.**

Question 6. Why is it not possible to have an ecosystem with only producers and no consumers or decomposers?
Answer: An ecosystem needs consumers to regulate producer populations (e.g., herbivores eat plants) and decomposers to recycle dead matter into nutrients for producers. Without consumers, producers would overgrow and die from competition; without decomposers, nutrients would not cycle, collapsing the system.

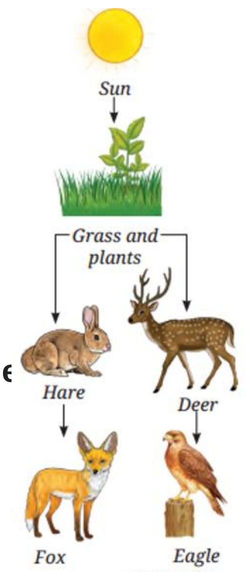
Question 7. Observe two different places near your home or school (e.g., a park and a roadside). List the living and non-living components you see. How are the two ecosystems different?
Answer: We can see some common things near our home and school, such as
- **Park: Living (trees, birds, squirrels, grass); Non-living (soil, water, benches, sunlight).**
- **Roadside: Living (weeds, insects); Non-living (asphalt, dust, car exhaust).**

Differences: The park is a designed ecosystem with diverse plants and animals, supported by soil and water, while the roadside is a disturbed, human-altered area with fewer species and more pollution.

Question 8: 'Human-made ecosystems like agricultural fields are necessary, but they must be made sustainable.' Comment on the statement.
Answer: Agree. Agricultural fields provide food but often use unsustainable practices (e.g., synthetic fertilisers, monoculture) that degrade soil and harm biodiversity. Sustainable methods (e.g., organic farming, crop rotation) ensure long-term productivity and environmental health.

Question 9.
If the Indian hare population (Fig.) drops because of a disease, how would it affect the number of other organisms?

Answer:
- **Predators: The numbers of foxes and eagles would decrease due to reduced prey (hares), leading to food scarcity.**
- **Plants: Grass and plant populations might increase initially without hare grazing, but overgrowth could alter the habitat.**
- **Deer: Since deer aren't preyed on by foxes or eagles, their role in this particular chain isn't disturbed directly.**
- **Why: A decline in a key herbivore like the hare disrupts the food web, affecting both higher (predators) and lower (plants) trophic le**



Parts of a food web